

Lesson 3 Teacher's Checklist / Prep Guide

Complete before class

VIDEO GAME CREATION

Self-Contained Lesson Assets

Per Lesson Production Standards Rule 3, nothing this lesson needs should require a student to search for or download something mid-class. Everything below is already prepared and included in this lesson's Assets folder - no download required.

- ☐ New placeholder projectile sprite already generated (bullet_yellow.png, 128x128 with transparency, same canvas size as Lesson 2's player sprites) - a simple yellow capsule shape, distinct from the round player sprites.
- ☐ New sound effect already generated (sfx_shoot.wav, a short descending "pew" - deliberately distinct from Lesson 2's sfx_blip.wav so shooting and moving don't sound the same) for the Level 3 "add a sound" idea.
- ☐ Lesson 2's Assets folder (player sprites, sfx_blip.wav) is still needed too, since this lesson builds on the same ongoing [YourName]Game project rather than starting fresh.
- ☐ Copy this lesson's Assets folder onto the shared classroom drive / USB backups alongside the handouts.

Continuity From Lesson 2

This is the first lesson where students open an existing project instead of starting a new one - flag this clearly at the start of lab, since it's a new habit.

- ☐ Remind students to open their own [YourName]Game project from Lesson 2, not create a new one - the shooting mechanic is added on top of the player object they already built.
- ☐ If a student's Lesson 2 project is lost or broken, have a pre-built "Lesson 2 complete" starter available per engine (from the Lesson 2 Teacher's Keys) so they aren't blocked.
- ☐ Note for students whose Lesson 2 player still moves in all 4 directions (the Level 2 stretch): that's fine to leave as-is, or trim to Left/Right only to match the actual capstone game - either is acceptable, since the capstone itself only calls for Left/Right (see the Lesson 2 GDD).

Engine Verification

- ☐ Confirm the demo laptop's Hello World and Lesson 2 exercise still work for all 8 engines - re-run any that haven't been touched in a few weeks.

- ☐ Pre-build the full Scratch shooting exercise (all levels) on top of a finished Lesson 2 project, so you have a working example ready to show.
- ☐ Godot: confirm a second PackedScene (the projectile) can be preloaded and instantiated from the player's script without errors in your installed version. Also confirm testing this week's exercise means pressing F5 (Run Project), not F6 on player.tscn alone - the bullet-spawn code relies on Player actually running as Main's child.
- ☐ Unity: confirm Instantiate() and a simple prefab workflow behave as expected in your installed Editor version.
- ☐ GameMaker: quick-test instance_create_layer() targeting the room's existing instance layer name in a throwaway object before class.
- ☐ GDevelop: confirm the "Destroy outside screen" behavior is available when adding a behavior to an object in your installed version (it should be pre-installed by default).
- ☐ TIC-80: confirm btnp() (press, not hold) is being used for firing, not btn() - btn() would fire a new bullet every single frame the key is held.
- ☐ 001 Game Creator (Windows demo laptop): confirm the "Map – Spawn Actor" and "Left/Entered Camera" Actor Trigger options are accessible in the Script/Actor editors.
- ☐ If any advanced-track students are on Unreal this week, confirm Spawn Actor from Class and a simple Bullet Blueprint compile cleanly.

Materials

- ☐ Print or queue digital copies of all 8 Student Handouts (or confirm students can access their engine's version electronically).
- ☐ Have the 8 Teacher's Keys on hand (instructor reference only - do not distribute) in case a student needs a worked example beyond Level 1.
- ☐ Bring: demo laptop, projector cable/adapter, handout copies, Assets folder (on USB as backup).

Known Gotchas This Week

- Every engine's Level 1 uses a "fire once per press" check (btnp(), "Pressed" pin, a boolean latch, etc.) rather than "fire every frame the key is held" - flag this early, since it's the single most common bug this week (a held key spawning dozens of projectiles per second).
- Scratch: creating and deleting a clone within the same "when I start as a clone" stack can freeze the project - the movement script and the off-screen cleanup check must be the same stack (a clone only runs one "when I start as a clone" script), but avoid also creating new clones from inside that same stack.
- GDevelop: the Destroy Outside Screen behavior only removes an object once it's fully past the deletion margin (200px default) - a projectile won't vanish the instant it touches the very edge of the screen, and that's expected, not a bug.
- 001 Game Creator: dialog field layouts and exact event names can vary slightly by version - if something looks different from the handout, the Script Editor and Actor wiki pages have the current layout.
- Unreal: this is the heaviest engine on our list - if a student's laptop is struggling, it's fine for them to observe this week's demo and build the exercise in a lighter engine instead.